

Bluetooth-Controlled Car Using UART RTL

RTL Design and Physical-Design Project Summary

This report documents the project architecture and points to the project's simulation and OpenLane reports. Physical hardware operation and numerical PPA results are not claimed unless supported by recorded evidence.

BLUETOOTH-CONTROLLED CAR USING UART RTL

PROJECT SUMMARY

PROJECT INFORMATION

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Project Name       : Bluetooth-Controlled Car Using UART RTL  
Top Module        : uart_bt_car_top  
Design Flow       : RTL-to-GDSII  
Physical Design   : OpenLane  
Technology        : Sky130
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DESIGN ARCHITECTURE

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RTL Modules       : uart_bt_car_top.v  
                  : bt_car_controller.v  
                  : uart_rx.v  
  
Communication     : UART serial input from Bluetooth module  
Command Mapping   : F - Forward  
                  : B - Backward  
                  : L - Left  
                  : R - Right  
                  : S - Stop  
  
Motor Interface   : Control outputs for an external motor driver  
Hardware Testing  : Not verified
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VERIFICATION

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RTL Simulation    : Refer to the simulation results and waveform  
VCD Waveform     : simulation/uart_bt_car.vcd
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PHYSICAL DESIGN

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Synthesis        : Refer to the actual OpenLane reports  
Floorplanning    : Refer to the actual OpenLane reports  
Placement        : Refer to the actual OpenLane reports  
Routing          : Refer to the actual OpenLane reports  
Signoff          : Refer to the actual DRC/LVS reports
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TIMING, AREA AND POWER

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Values are intentionally not listed here until verified from  
this Bluetooth car project's own OpenLane reports.
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